

interestingly though, in some cases we found notable differences between the results coming from our approach relative to the more common two-way fixed effects approach. These differences suggest that using an approach that is robust to treatment effect heterogeneity and dynamics should be strongly considered by applied researchers.

Appendix A: Proofs of Main Results

We provide the proofs of our results in this appendix. Before proceeding, we first state and prove several auxiliary lemmas that help us to prove our main theorems.

Let

$$ATT_X(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) | X, G_g = 1].$$

Lemma A.1. *Let Assumptions 1, 2, 3, 4, and 6 hold. Then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$,*

$$ATT_X(g, t) = \mathbb{E}[Y_t - Y_{g-\delta-1} | X, G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | X, C = 1] \text{ a.s..}$$

Proof of Lemma A.1: In what follows, take all equalities to hold almost surely (a.s.). Then, we have that

$$\begin{aligned} ATT_X(g, t) &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \mathbb{E}[Y_t(0) - Y_{g-\delta-1}(0) | X, G_g = 1] \\ &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \sum_{\ell=0}^{t-g-\delta} \mathbb{E}[\Delta Y_{t-\ell}(0) | X, G_g = 1] \\ &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \sum_{\ell=0}^{t-g-\delta} \mathbb{E}[\Delta Y_{t-\ell}(0) | X, C = 1] \\ &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \mathbb{E}[Y_t(0) - Y_{g-\delta-1}(0) | X, C = 1] \\ &= \mathbb{E}[Y_t - Y_{g-\delta-1} | X, G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | X, C = 1] \end{aligned}$$

where the first equality follows from adding and subtracting $\mathbb{E}[Y_{g-\delta-1}(0) | X, G_g = 1]$, the second equality from simple algebra, the third equality by Assumption 4, the fourth equality by simple algebra, and the last equality from (2.1) and Assumption 3. $\square$

Lemma A.2. *Let Assumptions 1, 2, 3, 5 and 6 hold. Then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ with $g - \delta \leq t < \bar{g}$*

$$ATT_X(g, t) = \mathbb{E}[Y_t - Y_{g-\delta-1} | X, G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | X, D_{t+\delta} = 0, G_g = 0] \text{ a.s..}$$

Proof of Lemma A.2: The proof follows similar steps as the proof of Lemma A.1. Taking all equalities to hold almost surely (a.s.), we have that

$$\begin{aligned} ATT_X(g, t) &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \mathbb{E}[Y_t(0) - Y_{g-\delta-1}(0) | X, G_g = 1] \\ &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \sum_{\ell=0}^{t-g-\delta} \mathbb{E}[\Delta Y_{t-\ell}(0) | X, G_g = 1] \\ &= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \sum_{\ell=0}^{t-g-\delta} \mathbb{E}[\Delta Y_{t-\ell}(0) | X, D_{t+\delta} = 0, G_g = 0] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E}[Y_t(g) - Y_{g-\delta-1}(0) | X, G_g = 1] - \mathbb{E}[Y_t(0) - Y_{g-\delta-1}(0) | X, D_{t+\delta} = 0, G_g = 0] \\
&= \mathbb{E}[Y_t - Y_{g-\delta-1} | X, G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | X, D_{t+\delta} = 0, G_g = 0]
\end{aligned}$$

where the first equality follows from adding and subtracting $\mathbb{E}[Y_{g-\delta-1}(0) | X, G_g = 1]$, the second equality from simple algebra, the third equality by Assumption 5 with $s = t + \delta$, the fourth equality by simple algebra, and the last equality from (2.1) and Assumption 3. $\square$

Now, we are ready to proceed with the proofs of our main theorems.

Proof of Theorem 1:

Part 1: Identification when Assumption 4 is invoked.

In this case, given the result in Lemma A.1,

$$\begin{aligned}
ATT(g, t) &= \mathbb{E}[ATT_X(g, t) | G_g = 1] \\
&= \mathbb{E}[(\mathbb{E}[Y_t - Y_{g-\delta-1} | X, G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | X, C = 1]) | G_g = 1] \\
&= \mathbb{E}[Y_t - Y_{g-\delta-1} | G_g = 1] - \mathbb{E}\left[\underbrace{\mathbb{E}[Y_t - Y_{g-\delta-1} | X, C = 1]}_{=m_{g,t,\delta}^{nev}(X)} | G_g = 1\right] \\
&= \mathbb{E}\left[\frac{G_g}{\mathbb{E}[G_g]} (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(X))\right].
\end{aligned}$$

Hence, we have that $ATT(g, t) = ATT_{or}^{nev}(g, t; \delta)$.

Next, to show that $ATT(g, t) = ATT_{ipw}^{nev}(g, t; \delta)$, it suffices to show that

$$\frac{\mathbb{E}\left[\frac{p_g(X)C}{(1-p_g(X))}(Y_t - Y_{g-\delta-1})\right]}{\mathbb{E}\left[\frac{p_g(X)C}{(1-p_g(X))}\right]} = \frac{\mathbb{E}[G_g \cdot \mathbb{E}[Y_t - Y_{g-\delta-1} | X, C = 1]]}{\mathbb{E}[G_g]}. \quad (\text{A.1})$$

Towards this end, by noticing that

$$p_g(X) = \frac{\mathbb{E}[G_g | X]}{\mathbb{E}[G_g + C | X]}, \quad 1 - p_g(X) = \frac{\mathbb{E}[C | X]}{\mathbb{E}[G_g + C | X]}, \quad (\text{A.2})$$

it follows that

$$\begin{aligned}
\mathbb{E}\left[\frac{p_g(X)C}{1-p_g(X)}\right] &= \mathbb{E}\left[\frac{\mathbb{E}[G_g | X]C}{\mathbb{E}[C | X]}\right] \\
&= \mathbb{E}\left[\frac{\mathbb{E}[G_g | X]\mathbb{E}[C | X]}{\mathbb{E}[C | X]}\right] \\
&= \mathbb{E}[\mathbb{E}[G_g | X]] \\
&= \mathbb{E}[G_g]. \quad (\text{A.3})
\end{aligned}$$

Next, by exploiting (A.2) and applying the law of iterated expectations, we have that

$$\begin{aligned}
\mathbb{E}\left[\frac{p_g(X)C}{(1-p_g(X))}(Y_t - Y_{g-\delta-1})\right] &= \mathbb{E}\left[\frac{\mathbb{E}[G_g | X]C}{\mathbb{E}[C | X]}(Y_t - Y_{g-\delta-1})\right] \\
&= \mathbb{E}\left[\frac{\mathbb{E}[G_g | X]}{\mathbb{E}[C | X]}\mathbb{E}[C \cdot (Y_t - Y_{g-\delta-1}) | X]\right] \\
&= \mathbb{E}[\mathbb{E}[G_g | X] \cdot \mathbb{E}[(Y_t - Y_{g-\delta-1}) | X, C = 1]]
\end{aligned}$$

$$= \mathbb{E} [G_g \cdot \mathbb{E} [(Y_t - Y_{g-\delta-1}) | X, C = 1]].$$

Thus, combined this with (A.3), we establish (A.1), implying that $ATT(g, t) = ATT_{ipw}^{nev}(g, t; \delta)$.

Finally, notice that

$$\begin{aligned} ATT_{dr}^{nev}(g, t; \delta) &= \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[\frac{p_g(X)C}{1-p_g(X)} \right]} \right) (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(X)) \right] \\ &= \mathbb{E} \left[\underbrace{\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[\frac{p_g(X)C}{1-p_g(X)} \right]} \right) (Y_t - Y_{g-\delta-1})}_{\equiv ATT_{ipw}^{nev}(g, t; \delta)} \right] \\ &\quad - \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[\frac{p_g(X)C}{1-p_g(X)} \right]} \right) m_{g,t,\delta}^{nev}(X) \right] \\ &= ATT(g, t) - \frac{1}{\mathbb{E}[G_g]} \mathbb{E} \left[\left(G_g - \frac{\mathbb{E}[G_g|X]C}{\mathbb{E}[C|X]} \right) m_{g,t,\delta}^{nev}(X) \right] \\ &= ATT(g, t) - \frac{1}{\mathbb{E}[G_g]} \mathbb{E} [(\mathbb{E}[G_g|X] - \mathbb{E}[G_g|X]) \cdot m_{g,t,\delta}^{nev}(X)] \\ &= ATT(g, t). \end{aligned}$$

where the third equality follows from $ATT_{ipw}^{nev}(g, t; \delta) = ATT(g, t)$, (A.2) and (A.3), and the fourth equality from the law of iterated expectations.

Part 2: Identification when Assumption 5 is invoked.

In this case, given the result in Lemma A.2,

$$\begin{aligned} ATT(g, t) &= \mathbb{E}[ATT_X(g, t) | G_g = 1] \\ &= \mathbb{E}[(\mathbb{E}[Y_t - Y_{g-\delta-1} | X, G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | X, D_{t+\delta} = 0, G_g = 0]) | G_g = 1] \\ &= \mathbb{E}[Y_t - Y_{g-\delta-1} | G_g = 1] - \mathbb{E} \left[\underbrace{\mathbb{E}[Y_t - Y_{g-\delta-1} | X, D_{t+\delta} = 0, G_g = 0]}_{=m_{g,t,\delta}^{ny}(X)} | G_g = 1 \right] \\ &= \mathbb{E} \left[\frac{G_g}{\mathbb{E}[G_g]} (Y_t - Y_{g-1-a} - m_{g,t,\delta}^{ny}(X)) \right]. \end{aligned}$$

Hence, we have that $ATT(g, t) = ATT_{dr}^{ny}(g, t; \delta)$.

Next, to show that $ATT(g, t) = ATT_{ipw}^{ny}(g, t; \delta)$, it suffices to show that

$$\frac{\mathbb{E} \left[\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)} (Y_t - Y_{g-\delta-1}) \right]}{\mathbb{E} \left[\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)} \right]} = \frac{\mathbb{E}[G_g \cdot \mathbb{E}[Y_t - Y_{g-\delta-1} | X, D_{t+\delta} = 0, G_g = 0]]}{\mathbb{E}[G_g]}. \quad (\text{A.4})$$

Towards this end, recall that $p_{g,t+\delta}(X) = P(G_g = 1 | X, G_g + (1 - D_{t+\delta})(1 - G_g) = 1)$ and also notice that

$$p_{g,t+\delta}(X) = \frac{\mathbb{E}[G_g | X]}{\mathbb{E}[G_g + (1 - D_{t+\delta})(1 - G_g) | X]}, \quad 1 - p_g(X) = \frac{\mathbb{E}[(1 - D_{t+\delta})(1 - G_g) | X]}{\mathbb{E}[G_g + (1 - D_{t+\delta})(1 - G_g) | X]}, \quad (\text{A.5})$$

it follows that by the law of iterated expectations,

$$\begin{aligned}\mathbb{E}\left[\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)}\right] &= \mathbb{E}\left[\frac{\mathbb{E}[G_g|X](1-D_{t+\delta})(1-G_g)}{\mathbb{E}[(1-D_{t+\delta})(1-G_g)|X]}\right] \\ &= \mathbb{E}[G_g].\end{aligned}\tag{A.6}$$

Next, by exploiting (A.5) and applying the law of iterated expectations, we have that

$$\begin{aligned}&\mathbb{E}\left[\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)}(Y_t - Y_{g-\delta-1})\right] \\ &= \mathbb{E}\left[\frac{\mathbb{E}[G_g|X](1-D_{t+\delta})(1-G_g)}{\mathbb{E}[(1-D_{t+\delta})(1-G_g)|X]}(Y_t - Y_{g-\delta-1})\right] \\ &= \mathbb{E}\left[\frac{\mathbb{E}[G_g|X]}{\mathbb{E}[(1-D_{t+\delta})(1-G_g)|X]}\mathbb{E}[(1-D_{t+\delta})(1-G_g) \cdot (Y_t - Y_{g-\delta-1})|X]\right] \\ &= \mathbb{E}[\mathbb{E}[G_g|X] \cdot \mathbb{E}[(Y_t - Y_{g-\delta-1})|X, D_{t+\delta} = 0, G_g = 0]] \\ &= \mathbb{E}[G_g \cdot \mathbb{E}[(Y_t - Y_{g-\delta-1})|X, D_{t+\delta} = 0, G_g = 0]].\end{aligned}$$

Thus, combined this with (A.6), we establish (A.4), implying that $ATT(g, t) = ATT_{ipw}^{ny}(g, t; \delta)$.

Finally, notice that

$$\begin{aligned}ATT_{dr}^{ny}(g, t; \delta) &= \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)}}{\mathbb{E}\left[\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)}\right]}\right)(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{ny}(X))\right] \\ &= ATT_{ipw}^{ny}(g, t; \delta) - \frac{1}{\mathbb{E}[G_g]}\mathbb{E}\left[\left(G_g - \frac{\mathbb{E}[G_g|X](1-D_{t+\delta})(1-G_g)}{\mathbb{E}[(1-D_{t+\delta})(1-G_g)|X]}\right)m_{g,t,\delta}^{ny}(X)\right] \\ &= ATT(g, t) - \frac{1}{\mathbb{E}[G_g]}\mathbb{E}[(\mathbb{E}[G_g|X] - \mathbb{E}[G_g]) \cdot m_{g,t,\delta}^{nev}(X)] \\ &= ATT(g, t).\end{aligned}$$

where the second equality follows from (A.5) and (A.6), and the third equality from $ATT_{ipw}^{nev}(g, t; \delta) = ATT(g, t)$ and the law of iterated expectations. $\square$

Proof of Theorem 2: From Theorem 1 it follows that $ATT(g, t)$'s are point-identified for all groups g and time periods t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$. For each (g, t) pair, the asymptotic linear representation of $\sqrt{n}(\widehat{ATT}_{dr}^{nev}(g, t; \delta) - ATT(g, t))$ follows from Theorem A.1(a) of Sant'Anna and Zhao (2020), whereas

$$\sqrt{n}(\widehat{ATT}_{t \geq (g-\delta)}^{dr, nev} - ATT_{t \geq (g-\delta)}) \xrightarrow{d} N(0, \mathbb{E}[\Psi_{t \geq (g-\delta)}^{dr, nev}(W)\Psi_{t \geq (g-\delta)}^{dr, nev}(W)'])$$

follows from the Lindeberg–Lévy central limit theorem. $\square$

Proof of Theorem 3: Note that, by the conditional multiplier central limit theorem, see Lemma 2.9.5 in van der Vaart and Wellner (1996), as $n \rightarrow \infty$,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \Psi_{t \geq (g-\delta)}^{dr, nev}(W_i) \xrightarrow{*} N(0, \Sigma),\tag{A.7}$$

where $\Sigma = \mathbb{E}[\Psi_{t \geq (g-\delta)}^{dr, nev}(W)\Psi_{t \geq (g-\delta)}^{dr, nev}(W)']$. Thus, to conclude the proof that

$$\sqrt{n} \left(\widehat{ATT}_{t \geq (g-\delta)}^{*, dr, nev} - \widehat{ATT}_{t \geq (g-\delta)}^{dr, nev} \right) \xrightarrow{*} N(0, \Sigma),$$